



NSC 4330

Technology and the Nervous System

Final Presentation | Aryamaan Seth | Spring 2026

What I set out to do

TOPIC — Technology and its impact on the nervous system

01

Improve public speaking through weekly paper presentations

02

Predict how AI will impact the structure and function of the nervous system

03

Generate ideas for systemic changes to address potential pitfalls of AI

04

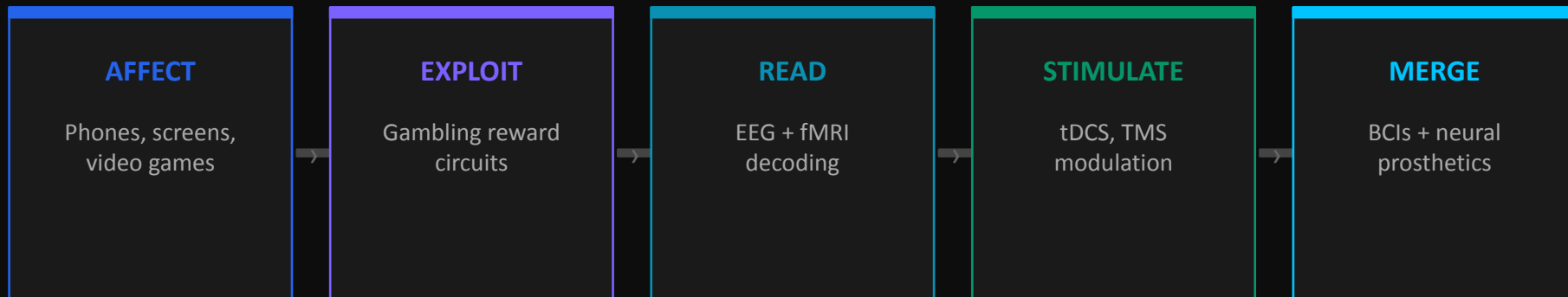
Generate ideas for personal behaviors that guard against AI-related harms

05

Explore what blending technology with the nervous system actually means

A single arc, from outside to inside

Every article was selected to trace technology moving closer to the brain. Starting with distraction and behavior, moving through reward exploitation and brain reading, and ending with direct brain integration.



The selection wasn't random. Each paper was chosen because it moved the needle one step closer to the brain itself.

What the literature actually says

Technology changes the brain at every level of contact

Screen time alters cortical thickness in toddlers. Video games expand attentional capacity. Gambling mechanics hijack dopamine prediction error. The brain does not stay neutral when exposed to technology.

The relationship is becoming bidirectional

Early papers showed tech acting on the brain. Later papers flipped that. Lorach, Kunz, and Valle show the brain now directs technology in real time. Thought controls limbs. Speech is decoded silently. Touch is written back in.

The boundary between tool and body is dissolving

A hippocampal prosthetic is not a tool you use. A brain-spine interface that restores walking is not a device you wear. The question is no longer whether tech will merge with the nervous system. It already has, in specific people, for specific functions.

The same problems kept showing up

Small N

Lorach: n=1. Kunz: n=4. Valle: n=2. Viñas-Guasch Study 2: n=14. Even better-powered studies like García-Fernández had 120 completers. The field is doing groundbreaking work with tiny samples.

Healthy Young Adults

Nearly every non-clinical paper recruited university students or healthy volunteers aged 18-30. Hutton was an exception. Most findings can't be assumed to generalize to older adults, clinical populations, or kids.

No Long-Term Follow-Up

How long do tDCS effects last? Does BCI use change the brain over years? Does childhood screen time produce lasting structural changes? Almost none of the papers tried to answer this.

Ethics Absent From the Science

Ozcelik reconstructed what people were looking at from their brain scans and published no ethical framework. Kunz confirmed BCIs leak unintended thoughts and devoted a few lines to it. The technology is moving faster than the conversation around it.

Where this is going next

01 Closed-loop BCIs at scale

Lorach, Kunz, and Valle each solved one piece. Motor output, speech input, tactile feedback. The next step is integrating all three into a single bidirectional system that talks to multiple brain regions simultaneously and in real time.

02 Personalized stimulation using individual neuroimaging

Both tDCS and TMS studies used fixed electrode placements for all participants regardless of individual anatomy. fMRI-guided, person-specific targeting is the obvious improvement. Viñas-Guasch already used this approach for TMS targeting. It needs to become the standard.

03 Legal and ethical frameworks for brain data

EEG can decode arousal. fMRI can reconstruct what you were looking at. BCIs leak unintended thoughts. None of these papers had a serious ethical framework. The science has moved so fast that no policy infrastructure exists yet for who owns neural data and what can be done with it.

04 Longitudinal population-level work on digital tech

Hutton and Chen showed early snapshots of screen time affecting brain structure. We need decade-scale cohort studies that track the same people from early childhood through adulthood to understand whether these changes are permanent, reversible, or adaptive.



The technology that started by distracting us is ending by becoming us.

Whether that is a tool or a threat depends entirely on the frameworks we build now, before the technology builds them for us.